Tomas Franco

Robotics & Mechatronics · Cal Poly SLO · M.S. Mechanical Engineering (expected June 2026)
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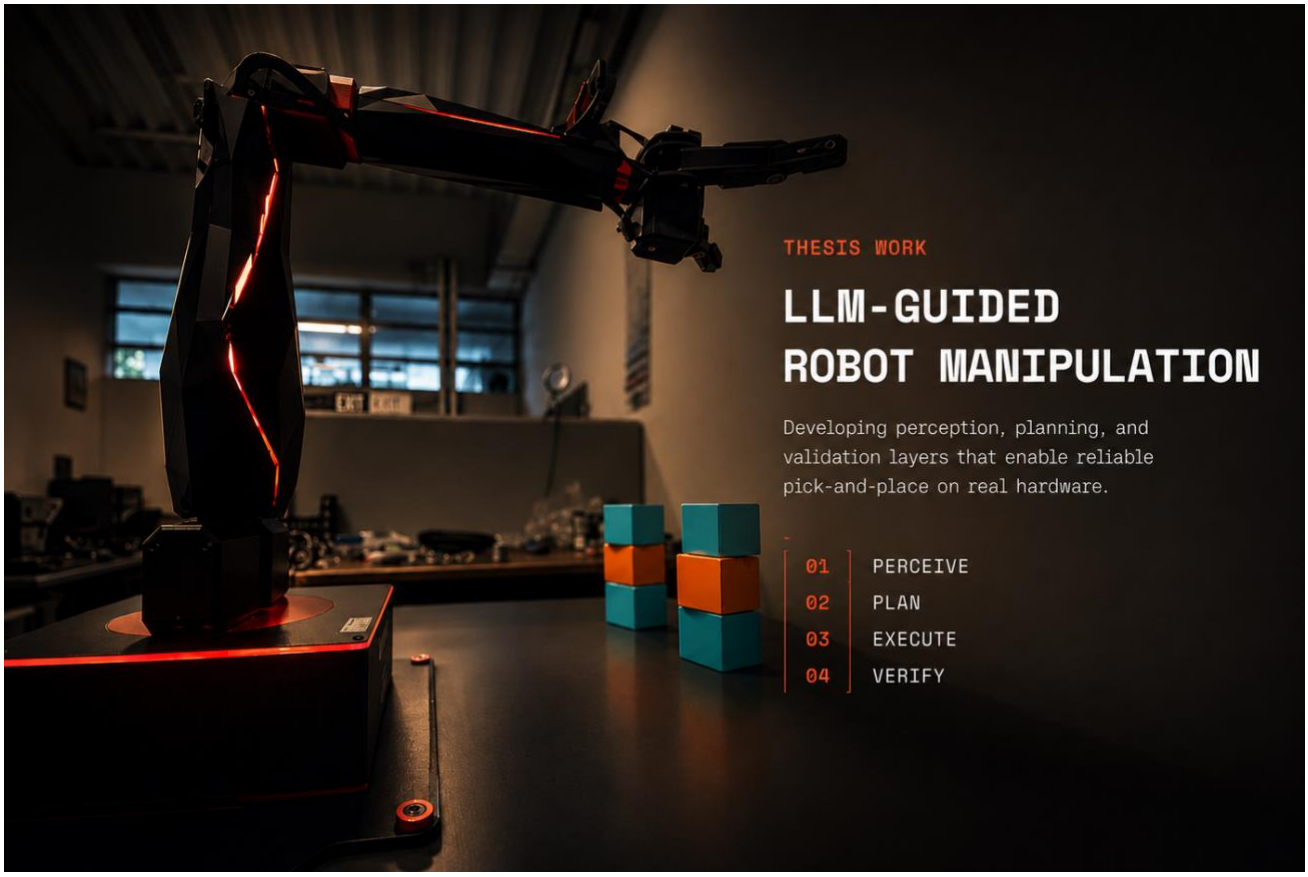
Thesis: LLM-led Autonomous Pick-and-Place Pipeline (In-Progress)

Video: https://www.youtube.com/watch?v=ye8d9MgdUjU



For my thesis, I'm developing a robust autonomous manipulation pipeline using RGB/depth sensing, current sensing, encoders, and YOLO instance segmentation. The platform I'm using is a 4 DOF Quanser Qarm, the same robot used in Healbot.

The primary tasks are cube color sorting and stacking based on text prompts. Initial grasping failures led to current-based grasp checks, integral and proportional gain tuning for smoother centering on detected objects, pre-grasp scene scans to avoid blindly placing cubes into occupied positions, and motion safety limits to keep the robot from getting too creative.

The broader goal is to evaluate how well a constrained LLM-guided manipulation pipeline can generalize across tabletop cube-sorting missions on limited-DOF hardware. Since LLMs can hallucinate or choose actions that are not physically valid, the robot is not allowed to rely on the planner alone. Instead, the system keeps the LLM grounded through vision-based state estimation, command validation, safety checks, and post-action verification before continuing to the next step.

Autonomous Driving Pipeline — MXCarkit

Video: youtube.com/shorts/HvuyGzoIKhY

For our final project in an autonomous vehicles course at Cal Poly, our team built a full autonomous driving pipeline on the MXCarkit platform — a small vehicle with an onboard computer and multi-sensor stack. The goal was smooth lane following with traffic-light awareness.

We built two parallel pipelines in Docker: a primary behavioral cloning model (PilotNet) mapping camera images directly to steering and speed commands, and a YOLOv11 backup for traffic light detection as a safety net. We collected driving data with Foxglove, ran 10 training trials across different architectures and hyperparameters tracked in Weights and Biases, and compared RGB vs. grayscale inputs. The color variant won — grayscale introduced lag and overshoot during turns.

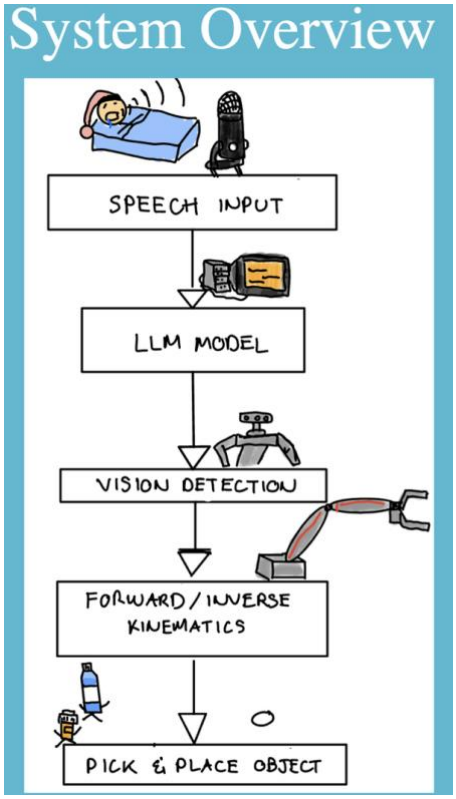
Our best model hit strong steering accuracy and the YOLO backup reached 0.98 F1 across red, yellow, and green classes. The YOLO model was exported to ONNX and integrated into a ROS2 node that published speed commands in response to detected light states. A great end-to-end experience: raw data collection → training → simulation validation → live hardware deployment.

Healbot

GitHub: github.com/TomasEzFranco/Healbot · **Video:** youtube.com/watch?v=xKzFN9P7FqU

Healbot is a robotic assistant to help people with mobility limitations retrieve important items — water, medicine, or an energy drink — based on their symptoms or request. If a patient says they are thirsty, the robot will quickly retrieve water, as long as we remembered to stock the very sophisticated robot pantry.

Built on the Quanser Qarm educational robot platform, the pipeline takes speech input, processes it through an LLM, uses YOLO for object detection, and performs pick-and-place operations at different waypoints using forward and inverse kinematics.

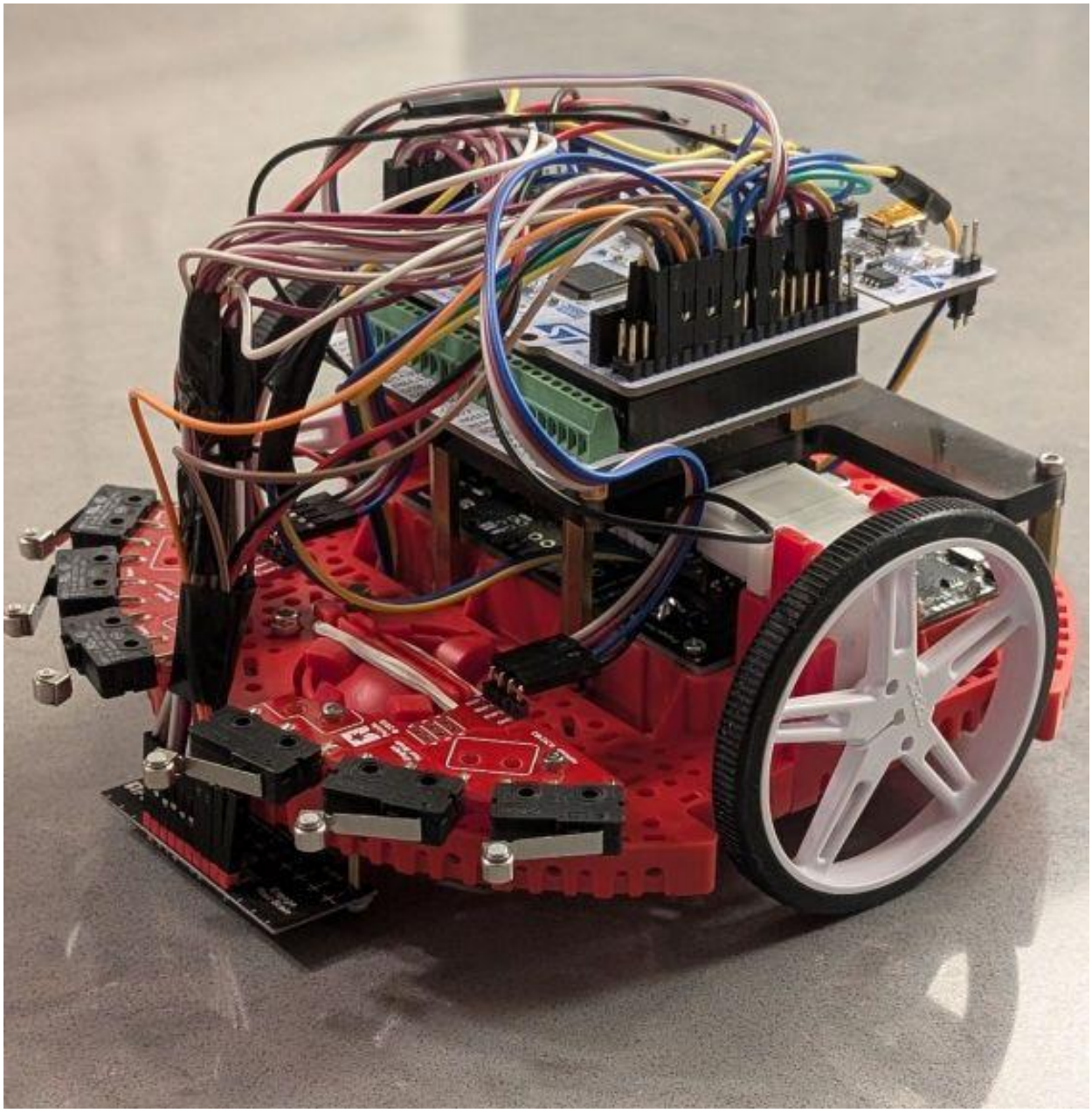


ROMI — Line Following Robot

GitHub: github.com/TomasEzFranco/Romi_Line_Follower · **Video:** youtube.com/watch?v=yved8C4mFfU

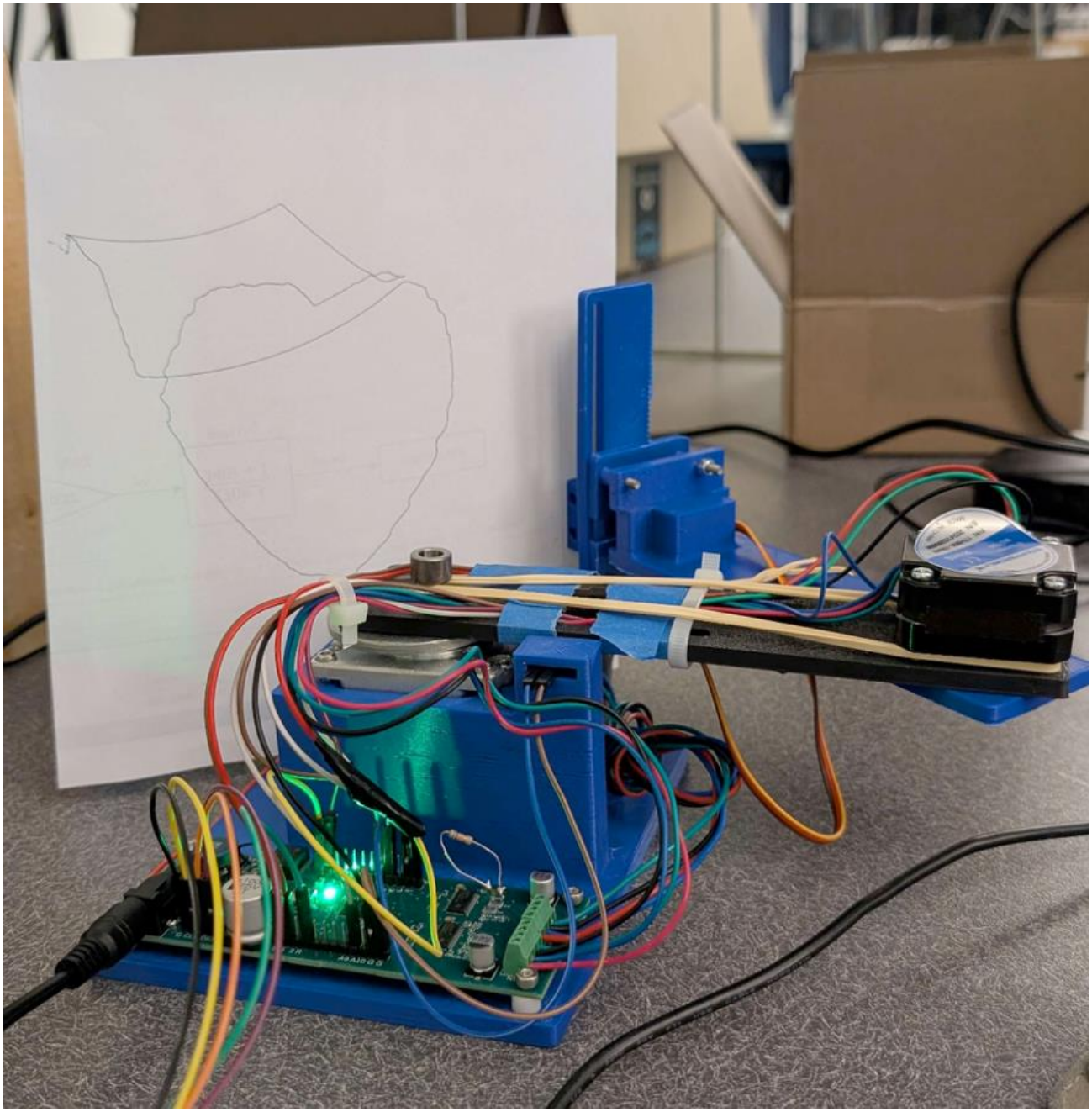
ROMI is a line-following differential drive robot that had to navigate a game track, hit checkpoints, and detect a wall without getting lost. We handled part selection, assembly, wiring diagrams, schematics, sensor integration, and firmware.

ROMULUS, as we called it, used an IR sensor array for line following, encoders for tracking wheel motion, and a BNO055 IMU for the parts of the track where the line disappeared, and the robot had to believe in itself. We characterized the motors and used closed-loop control with IR, encoder, and IMU feedback rather than open-loop moves. The final system combined line following, heading control, and dead reckoning with bump sensing to complete the course consistently.



DAL-I — Robotic Drafter

GitHub: github.com/TomasEzFranco/Robot-Aided-Drafter

DAL-I is our drafting robot. Unlike most plotters which follow a 2P or 3P design, DAL-I uses an RRP robot architecture introducing difficulty with nonlinear plotting and unnatural positions.

We designed the mechanical assembly, PCB, wiring diagrams, and software needed to bring DAL-I to life. My work focused on electrical design, schematics, wiring diagrams and board bring-up, and software development. While DAL-I's art may not make it to the Louvre anytime soon, it was a great experience taking a project from idea to a working physical system and we learned simple is often best.